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Stability Selection in Logistic Regression

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1. Objectives

Aim of the project

- Understand model building mechanisms; introduction to stability selection
- Evaluate and compare performance of model building methods ; Stability vs rest of the methods; model selection simulation

2. Problem Definition

Growing up in Ireland Study

- Growing Up in Ireland Survey ;Ireland’s first longitudinal study of school children,
- Two waves- Child Cohort and Infant Cohort
- Starting point for our project; 6359 observations, Logistic regression: predictors and response variables

3. Methodology

Logistic Regression

- Fitting a linear regression model to binary response variables

$$\text{Logit}(p_i) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots\ldots$$

Linear Predictor

Log Odds Probability of success of response variable

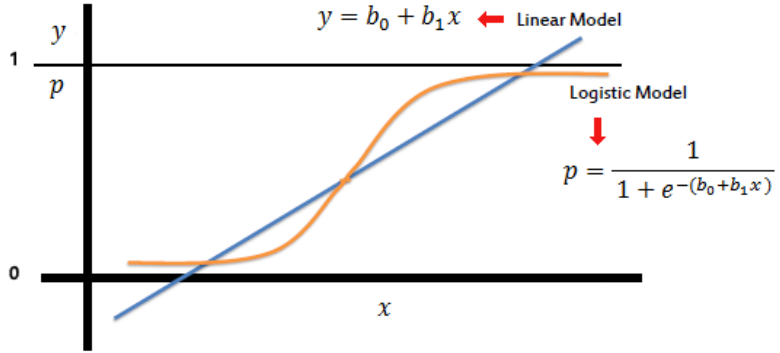


Fig 1 Sigmoid vs Straight Line Fit

- Sigmoidal response distribution → Link Function for transformation

3. Methodology (Contd…)

Modelling problem with Logistic Regression

Variables	Status
Secondary School student's engagement with Mathematics	Binary response
Student's transition difficulty	Categorical predictor
Attitude towards primary school	Categorical predictor
Attitude towards primary level Mathematics	Categorical predictor
Positive/Negative interaction with teachers	Categorical predictor
Gender of student	Categorical predictor
Current year in secondary school	Categorical predictor
Highest qualification student seeks to attain	Categorical predictor
Primary School student's engagement with Mathematics	Categorical predictor

3. Methodology (Contd…)

Traditional Model Building Methods

- Likelihood Ratio Test (LRT)
  - Compare Nested Models likelihoods
  - Likelihood Ratio Statistic (Chi Square) for significance of variables
- Stepwise Variable selection
  - Using LRT's repeatedly (Forward and Backward)
  - Akaike Information Criteria (Deviance + penalty for number of parameters)

3. Methodology (Contd…)

Traditional Model Building Methods

- Lasso Variable Selection
  - L1 Regularization approach

$$\hat{\beta}_\lambda = \arg min_{\beta}(-\ell(\beta) + \lambda \sum_{j=1}^{p} |\beta_j|)$$

Likelihood Function

L1 Regularization

  - λ estimation : K Fold cross validation

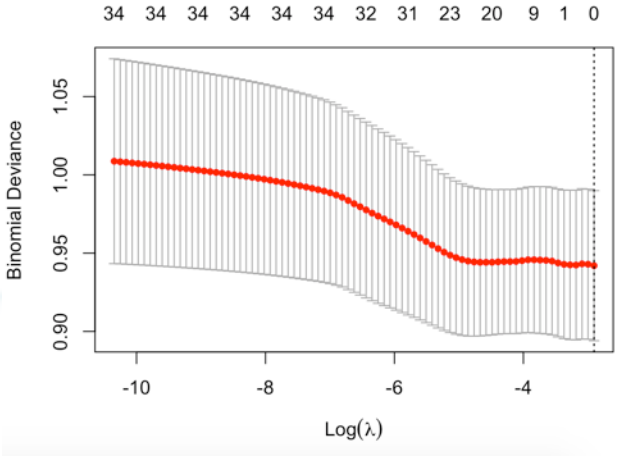


Fig 2 A CV Plot

3. Methodology (Contd…)

Advanced Model Building Methods

- Stability Selection
  - Subsampling approach : computing probability of feature selection from perturbed dataset
  - Can be wrapped around any variable selection method
  - More stable (consistent) results; lower type 1 error rate

3. Methodology (Contd…)

Design Parameters for Model Simulation

- Design Parameters
  - Performance Measures: Selection frequency of each predictor & selection frequency of number of predictors
  - True & ‘Noise’ Predictors: Additional ‘Noise’ included ‘Noise’ variables and Normal Random variates
  - Sample size: 500 & 5000 sized samples
  - Scenarios : 1) True predictors alone
    2) True predictors + ‘Noise’ predictors
    3) True predictors + Normal Random variates

4. Results

Results for Model Simulation

- Selection frequency of each predictor

Sample Size	Noise	Scenario	Result
500 & 5000	No	'True' predictors alone	Lasso > Stability > Stepwise > LRT
	Yes	+ 'Noise' predictors	Stability > Stepwise > LRT > Lasso
	Yes	+ 'Noise' Normal R.V	Stability > Stepwise > LRT > Lasso

- No considerable effect of sample size on the selection frequency of predictors

4. Results (Contd…)

Results for Model Simulation

– Selection frequency of number of predictors

Sample Size	Noise	Scenario	Result
500	No	'True' predictors alone	Lasso > Stability > Stepwise > LRT
	Yes	+ 'Noise' predictors	Stability > Stepwise > LRT > Lasso
	Yes	+ 'Noise' Normal R.V	Stepwise > LRT > Stability = Lasso
5000	No	'True' predictors alone	Stability = Lasso > Stepwise = LRT
	Yes	+ 'Noise' predictors	Stability > Stepwise = Lasso = LRT
	Yes	+ 'Noise' Normal R.V	Stability > Stepwise > LRT > Lasso

5. Conclusion & Synthesis

Takeaway from research

- Overall, Stability selection performed the best for both performance measures
- Stepwise and LRT did reasonably better against ‘noise’ predictors
- Stability failed to improve Lasso in cases where extraneous random variation was absent (true predictors alone) → Further research ?
- Stability selection’s overtly ‘inclusive’ nature → Further research ?
- Redundant step in stability selection: selecting random lambda values → lowest lambda values nonetheless selected. → Further research ?

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Questions ?